CHAPTER 1: Digital Systems and Binary Numbers

Answer all the following questions

1. Convert the following binary numbers into their decimal format:
    a) 101000
    b) 11111100
    c) 11100011
2. Convert the following decimal numbers into their binary equivalent:
    a) 17
    b) 66
    c) 46
3. Add the following binary numbers:
    a) 1010 + 0100
    b) 1111011 + 1100
4. Perform subtraction on the given binary numbers using the 2's complement. Check the result by converting to decimal
    a) 10011 - 10010
    b) 100010 - 100110
    c) 101000 – 10101
5. Convert the following hexadecimal numbers into their binary equivalent:
    a) 79
    b) ABC9
    c) 89EF
6. Convert the following binary numbers to its hexadecimal equivalent
    a) 1001111101010
    b) 11111000101001
7. Represent the following decimal numbers in 2's complement form.
    a) 45
    b) -60
    c) -120
8. Add the following signed numbers and detect if there is an over flow or not.
    a) 11111111 + 10111000
    b) 01011010 + 01110011
9. Find the decimal equivalent of the following signed binary numbers (2's complement form):
    a) 11111111
    b) 10000000
    c) 11010101

10. Convert the following decimal numbers to BCD:
    a) 104
    b) 185
    c) 210

11. Convert each of the BCD numbers to decimal:
    a) 1000 0000
    b) 0010 0011 0111
    c) 0110 0110 0110 0111